

MLOps

In-Class Assignment 6

1. A subscription company wants to build a model on **January 1, 2026** predicting whether a customer will churn during the next **90 days**, ending **March 31, 2026**. Churn is defined as having no active subscription on March 31.
 - a. Describe how the target would be constructed
 - b. Give some examples of x's in the data that could result in target leakage.
 - c. Let's consider what the loss is. What customers do you want to prioritize?
Answer in the context of the decisions to be made from the model.
2. Let's talk about delivery delays. The model is to be run before shipment (how much before?).
 - a. Based on the history of shipment data, describe
 - i. A possible binary target
 - ii. A possible multi-valued discrete target
 - iii. A possible continuous target
 - b. What useful data would be known from this standpoint in time? What would not be known and would thus constitute target leakage?
3. A hospital pharmacy makes its weekly medication order every **Monday at 8:00 AM**. On Monday, **November 2, 2026**, it must predict demand for a critical medication for November 2–8. The pharmacy estimates:
 - Each unused dose at the end of the week costs **\$18** because it expires.
 - Each dose of unmet demand costs **\$250** because an emergency replacement must be obtained.
 - Actual weekly demand in four historical weeks was 480, 510, 550, and 620 doses.
 - The model predicted 520 doses for each of those weeks.
 - a. What is the loss for each week?
 - b. Instead of 520, what constant prediction level would minimize the costs?
This can point to a data transformation to do.